

Study Notes: Introduction to Java

Programming ### 1. Overview of Java *

****Definition:**** A class-based, object-oriented, high-level programming language.

* ****Origin:**** Invented by ****James Gosling**** and ****Sun Microsystems**** in 1991.

* ****Naming History:**** Originally named ***Oak***, then ***Green***, then ***Java Coffee***, and finally ****Java**** (1995).

****Core Philosophy:**** "Write Once, Run Anywhere" (WORA).

2. Java Editions

Java provides three specialized editions based on the application type: *

- **Java Standard Edition (SE):**** Includes base libraries (e.g., `'java.lang'`, `'java.util'`).

- **Java Enterprise Edition (EE):**** Used for large-scale enterprise applications; includes APIs like JMS, EJB, JSP, and Servlets.

- **Java Micro Edition (ME):**** Designed for mobile devices, set-top boxes, and handheld gadgets.

3. Key Architecture & Execution *

****Compilation:**** Java source code is converted into ****Bytecode**** by a compiler.

* ****Execution:**** The ****Java Virtual Machine (JVM)**** interprets the bytecode, providing a platform-independent runtime environment.

* ****Comparison:**** Java is based on C and C++, but improved/simplified to eliminate common programming errors.

4.

Fundamental Concepts * **Methods:**

Blocks of code that perform a specific task when called (e.g., adding numbers). *

Classes: A collection of methods used to organize code (similar to categorized sections in a supermarket). ### 5. Key

Features of Java * **Platform

Independent:** Can run on any machine thanks to bytecode. * **Robust:**

Features **garbage collection** (memory management) and **exception handling**.

* **Secure:** Highly secure and virus-free.

* **Multi-threading:** Allows for multitasking within a program. *

Portable: Can run on any platform; data types have fixed sizes. *

Performance: Slightly slower than C/C++, but offers high performance compared to most other languages. ### 6.

OOP Concepts (Object-Oriented Programming)

Java relies on four main pillars to make code easy to understand and maintain: 1.

Abstraction: Hiding complex background details and only showing relevant information to the user. 2.

Encapsulation: Securing data by wrapping it inside classes, restricting direct external access. 3.

Inheritance: Allowing one class to acquire the properties and behaviors (attributes) of another, which saves

coding time. 4. **Polymorphism:** The ability of a single method or object to perform different tasks in different

contexts. **7. Applications of Java**

Web & Mobile: Online banking,

shopping, and mobile apps. **Software**

Development: Desktop and console

applications. **Game Development.**

Embedded Systems: Used in electronic hardware like washing machines, TVs, and

air conditioners. --- **Exam Tip:**

Remember the acronym **"PIE-A"** for the

four OOP concepts (Polymorphism,

Inheritance, Encapsulation, Abstraction)

to help you recall them during your test!